Chapter 9, Part 1: Parametrics

Parametric Equations

Formula for the derivative of parametric equations:

- First derivative

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

This makes sense if we think about canceling dt

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$\left(\frac{dx}{dt} \neq 0 \right)$

- Second Derivative

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$$

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$\left(\frac{dx}{dt} \neq 0 \right)$

Examples

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$:

$x = t^2 + 1 \quad y = 4t^3 - 2t^2$

Finding first derivative:

$\frac{dx}{dt} = 2t$

$\frac{dy}{dt} = 12t^2 - 4t$

$\frac{dy}{dx} = \frac{12t^2 - 4t}{2t}$ **=6t-2**

Finding the second derivative

$$\frac{d}{dt}\left[\frac{dy}{dx}\right]=6$$
$$\frac{dx}{dt}=2t+3$$

$$\frac{d^2y}{dx^2}=\frac{6}{2t+3}$$

Given x=2cost and y=4sin2t, find $\frac{dy}{dx}$

$$\frac{dx}{dt}=-2sint$$
$$\frac{dy}{dt}=8cos2t$$
$$\frac{dy}{dx}=\frac{-2sint}{8cos2t}=\frac{-sint}{4cos2t}$$

Practice Problems:

1. **Find the** $\frac{dy}{dx}$ **and** $\frac{d^2y}{dx^2}$ **given x=lnx and y=**x^3+2x
2. **Find the** $\frac{dy}{dx}$ **and** $\frac{d^2y}{dx^2}$ **given x=2sint and y=5cost+3**

Parametric Equations (continued)
More formulas and definitions:

x'(t)= $\frac{dx}{dt}$ is the <u>rate at which the x coordinate is changing</u> with respect to t, or the <u>velocity of the particle in the horizontal</u> direction (left and right)

y'(t)= $\frac{dy}{dt}$ is the <u>rate at which the y coordinate is changing</u> with respect to t, or the <u>velocity of the particle in the vertical</u> direction (up and down)

- A particle is at rest when both $\frac{dx}{dt}$ and $\frac{dy}{dt}$ are equal to 0
- $\frac{dy}{dx}$ is the slope of the tangent line

Vectors:
s = <x(t) , y(t)> is the position at any time t
v = <x'(t) , y'(t)> is the velocity vector at any time t
a = <x"(t) , y"(t)> is the acceleration vector at any time t

[vectors can be contained within <> or ()]

Formula for <u>arc length</u> AND <u>distance travelled by particle</u>

Arc Length in Parametric Form

If a smooth curve does not intersect itself on an interval [a, b] (except possibly at the endpoints), then the arc length of the curve over the interval is:

$$\boxed{L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt}$$

This formula can be derived from

$$L = \int_c^d \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

Formula for the speed of the particle or the magnitude (length) of the vector

$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

Formula for particle displacement:

$$< \int_a^b x'(t)dt, \int_a^b y'(t)dt >$$

Example

Given: $x(t) = 2t^3 + 6t^2$ **and** $y(t) = t^4 + 3t^3$

Write the position vector

$< 2t^3 + 6t^2 , t^4 + 3t^3 >$

Find the velocity vector at t = 1

velocity vector: $< 6t^2 + 12t , 4t^3 + 9t^2 >$

at t = 1 : <18,13>

Find the acceleration vector at t = 2

velocity vector: $< 12t + 12 , 12t^2 + 18t >$

at t = 2 : <36,84>

Find the speed of the vector at t = 2

$\frac{dx}{dt} = 6t^2 + 12t$

$\frac{dy}{dt} = 4t^3 + 9t^2$

$$\sqrt{(6(2)^2 + 12(2))^2 + (4(2)^3 + 9(2)^2)^2} = 83.235$$

Find the distance travelled by the particle from t=0 to t=2

$$\text{Distance} = \int_0^2 \sqrt{(6t^2 + 12t)^2 + (4t^3 + 9t^2)^2}\, dt = 57.376$$

Practice Problems:

1. If the position vector is $< 4t^2$, ln(x)>

   Find the velocity and acceleration vectors

   Find the magnitude of the velocity vector at t=3

2. If $\frac{dx}{dt} = \sqrt{t^4 + 8}$ and $\frac{dy}{dt} = 4e^t - 2e^{-t}$
   - Find the speed of the particle and its acceleration vector at time t=2
   - Find the total distance travelled b the particle over the time interval $0 \leq t \leq 4$

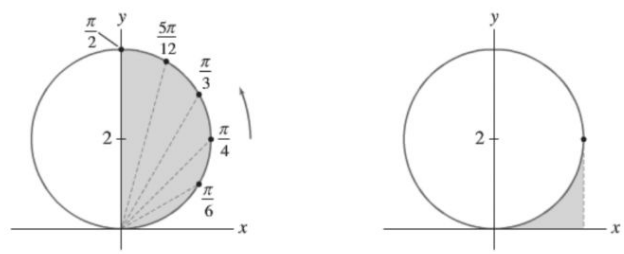
Chapter 9, Part 2: Polar

It is essential for this unit that you understand polar equations, what they look like, and how to sketch them. Please refer to the Precalc Polar page for a refresher!

Before we begin, a quick tip for sketching graphs is to plug in 0, $\pi/2$, π, and $3\pi/2$ into the equation for a general guideline and find where the equations equal 0 (AKA where the curve returns to the origin). For each problem, knowing the type of equation, where it equals 0, and where it is at each axis should be enough information to sketch the equation.

Polar Area

The concept for finding a specific area within a polar equation is quite similar to regular integration. Whereas ordinary integration finds the area underneath a curve, polar integration finds the area within a curve. And, whereas regular integration is between an interval of either x or y limits, polar integration is bound by radian limits.



(A) The polar integral computes the area swept out by a radial segment.

(B) The ordinary integral in rectangular coordinates computes the area underneath a curve.

You should know from prior knowledge that the area of a sector is $A = \frac{1}{2}\theta r^2$. The radius is a function of θ, thus the integral to solve for polar area looks like $\frac{1}{2}\int_a^b f(\theta)^2\, d\theta$. Looking at the image above, you can see that the equation can be represented by $r = 4sin(\theta)$. If you wanted to find the area of the shaded gray area, you would use the integral $\frac{1}{2}\int_0^{\pi/2} (4sin\theta)^2\, d\theta$. Makes sense, right?

Let's take a look at some examples of basic polar area questions.

Example 1: Find the area of one petal of r = 2sin(3θ).

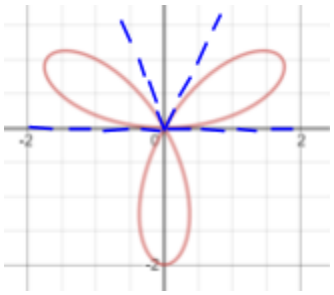
The first thing you should do is find the zeros of the equation. Since you should know approximately what this rose curve will look like, you'll only need the zeros for this problem.

$2\sin(3\theta) = 0$

$\sin(3\theta) = 0$

$3\theta = 0,\ \pi,\ 2\pi,\ 3\pi$

$\theta = 0,\ \pi/3,\ 2\pi/3,\ \pi$



In order to find the area of one petal, simply go from one zero to the next, which covers the entirety of the petal. Any of these could all be your final integrals:

$$\frac{1}{2}\int_{0}^{\pi/3} (2sin(3))^2\, d\theta$$

$$\frac{1}{2}\int_{\pi/3}^{2\pi/3} (2sin(3))^2\, d\theta$$

$$\frac{1}{2}\int_{2\pi/3}^{\pi} (2sin(3))^2\, d\theta$$

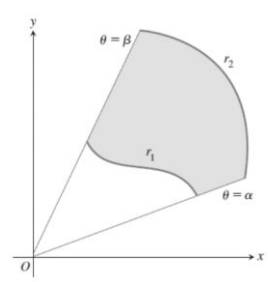
Be aware of what intervals actually represent what part of an equation. In this example, the petal on the bottom is actually on the interval $\pi/3$ to $2\pi/3$, although it doesn't look like it. Remember that it is on the bottom because the radius is negative during that interval, *but* it is still on that interval.

Example 2: Find the area of the inner loop of $r = 2 + 4\cos\theta$.

<u>Area Between Polar Curves</u>

To find the area between two polar curves is also quite similar to finding the area between two regular curves. The limits are either given to you, or they are where the curves intersect.

You can see how that checks out from the image on the right. If you find the area of the outer curve from the interval α to β and subtract the area of the inner curve from the interval α to β, then you have yourself the area of the in-between!

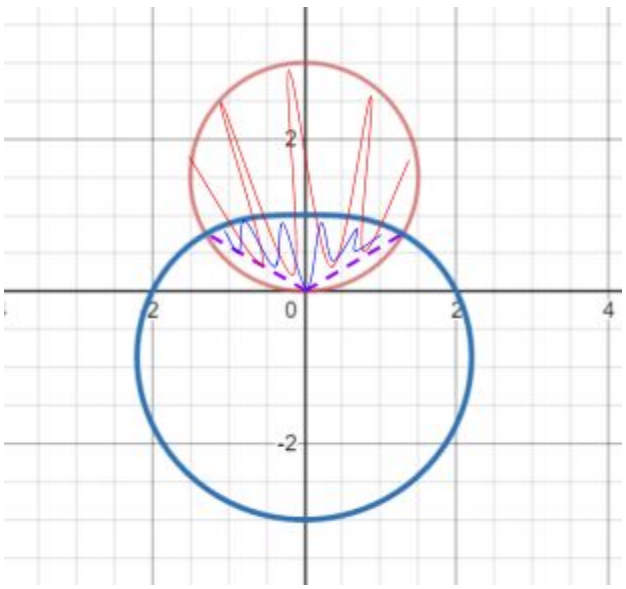


Area Between Polar Curves

The area of the region between $r_1(\theta)$ and $r_2(\theta)$ for $\alpha \le \theta \le \beta$ is

$$A = \int_\alpha^\beta \frac{1}{2} r_2^2 \, d\theta - \int_\alpha^\beta \frac{1}{2} r_1^2 \, d\theta = \int_\alpha^\beta \frac{1}{2}(r_2^2 - r_1^2) \, d\theta.$$

Example 1: Find the area of the region that is inside r = 3sinθ and outside r = 2 - sinθ.



We want to find the region that is inside the red circle but outside the blue curve. In order to do so, we will find the area of where the red scribble is and subtract where the blue scribble is. Now all we have to do is find where the two curves intersect.

$3\sin\theta = 2 - \sin\theta$

$4\sin\theta = 2$

$\sin\theta = ½$

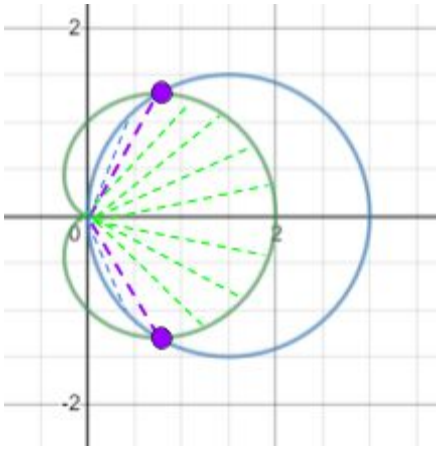
$\theta = \pi/6,\ 5\pi/6$

The final integral is this:

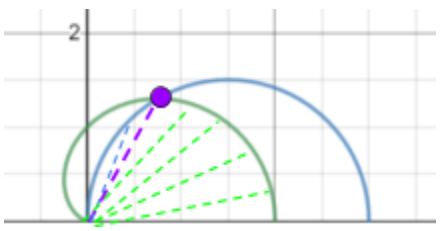
$$\frac{1}{2}\int_{\pi/6}^{5\pi/6} (3\sin\theta)^2 d\theta - \frac{1}{2}\int_{\pi/6}^{5\pi/6} (2 - \sin\theta)^2 d\theta$$

It can get a little harder with more complex problems, though. Let's walk through this example to see how you can break down harder problems:

Example 2: Find the area of the common interior of $r = 3cos\theta$ and $r = 1 + cos\theta$.



First things first: While it's not necessary, we're going to utilize symmetry to solve this problem. Since we know that the top half of the x axis matches the bottom half in this case, we're going to focus on finding the area for the common interior in the first quadrant and multiply the final product by 2.



Second: Now that we've got just the first half, we can see that the desired area is actually split into two, divided by the radian where they intersect. From 0 until the intersection, the curve we'll be integrating is the green cardioid. From the intersection until $\pi/2$ where the circle goes to 0, the curve we'll be integrating is the blue circle.

So, here's the final verdict:

$$2\left[\ \frac{1}{2}\int_{0}^{\pi/3}(1+cos\theta)^2 d\theta + \frac{1}{2}\int_{\pi/3}^{\pi/2}(3cos\theta)^2 d\theta\ \right]$$

Sidenote: If you wanted to solve for the area without using symmetry, know that your bottom limit should always be less than your upper limit. Area is always positive. To find that centerpiece with the green cardioid, your integral would go from $-\pi/3$ to $\pi/3$. Alternatively, you could make it go from $5\pi/3$ to $7\pi/3$. See how that works?

Polar Derivatives

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

$x = r\cos\theta$

$y = r\sin\theta$

Polar derivatives aren't hard, just... long. And tedious. As long as you remember the rules above, you should be set.

The steps to finding a derivative is simple. You write x and y equations by plugging in the radius equation into $r\cos\theta$ and $r\sin\theta$, respectively. Then you derive them and put $dy/d\theta$ over $dx/d\theta$ to find dy/dx.

Here's an example:

Find the equation of the line tangent to the polar curve $r = \sin(2\theta)$ when $\theta = 3\pi/4$.

$x = r\cos\theta = \sin(2\theta)\cos\theta$

$\frac{dx}{d\theta} = -\sin\theta\sin(2\theta) + 2\cos(2\theta)\cos\theta$

$y = r\sin\theta = \sin(2\theta)\sin\theta$

$\frac{dy}{d\theta} = \sin(2\theta)\cos\theta + 2\cos(2\theta)\sin\theta$

$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\sin(2\theta)\cos\theta+2\cos(2\theta)\sin\theta}{-\sin\theta\sin(2\theta)+2\cos(2\theta)\cos\theta}$

$\frac{dy}{dx}|_{\theta=3\pi/4} = 1$

$x(3\pi/4) = \frac{\sqrt{2}}{2}$

$y(3\pi/4) = -\frac{\sqrt{2}}{2}$

$y + \frac{\sqrt{2}}{2} = 1(x - \frac{\sqrt{2}}{2})$ is the final answer.

That's really about all there is to it! It's tedious, but it's just derivatives.

Solve these practice problems to check your understanding:

Sketch a graph and find the intersections without using a calculator. Use a calculator once you have your final integral. Find the area of:

1. One petal of $r = 4sin(4\theta)$
2. Interior of $r = 2 + 2cos(\theta)$
3. Inner loop of $r = 1 + 2cos(\theta)$
4. Common interior of $r = 4sin(\theta)$ and r = 2
5. Inside r = 2 and outside $r = 2 - sin(\theta)$

(Calcular Inactive)

1. Find the slope of the line tangent to the polar curve $r = 2 + 3sin(\theta)$ at $\theta = \pi$.
2. Find the equation of the line tangent to the polar curve $r = 2(1 - sin(\theta))$ at $\theta = \frac{7\pi}{6}$.
3. Find the values of θ in the interval $0 \leq \theta \leq 2\pi$ where the curve $r = 1 - sin(\theta)$ has a horizontal or vertical tangent.